

6.6 Classwork: The Discriminant and Polynomial Equations (no calculator)

1. [Maximum mark: 5]

For each quadratic function, find the discriminant $\Delta = b^2 - 4ac$ and state the number of real roots.

$$(a) \quad f(x) = x^2 - 6x + 9 \qquad [2]$$

$$(b) \quad g(x) = 2x^2 + 3x + 5 \quad [2]$$

(c) State what the discriminant tells you about the graph of each function. [1]

2. [Maximum mark: 6]

The equation $x^2 + kx + 9 = 0$ has two equal roots.

(a) Write down the discriminant in terms of k . [1]

(b) Find the values of k . [2]

(c) For each value of k , find the repeated root. [2]

(d) State the values of k for which the equation has no real roots. [1]

3. [Maximum mark: 7]

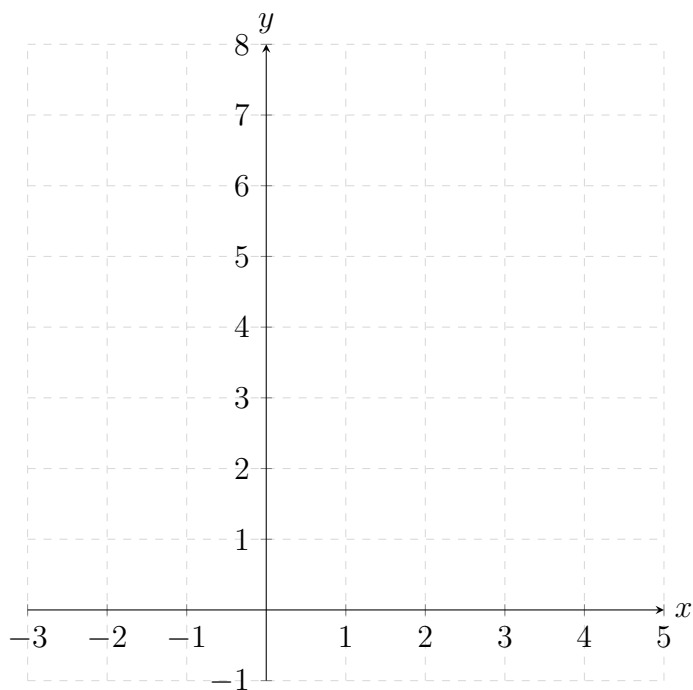
The line $y = 2x + c$ is tangent to the parabola $y = x^2 + 1$.

(a) Show that the x -coordinates of any intersection points satisfy $x^2 - 2x + (1 - c) = 0$.
[1]

(b) For the line to be tangent, the discriminant must equal zero. Find the value of c .
[2]

(c) Find the coordinates of the point of tangency. [2]

(d) On the axes below, sketch the parabola and the tangent line. [2]



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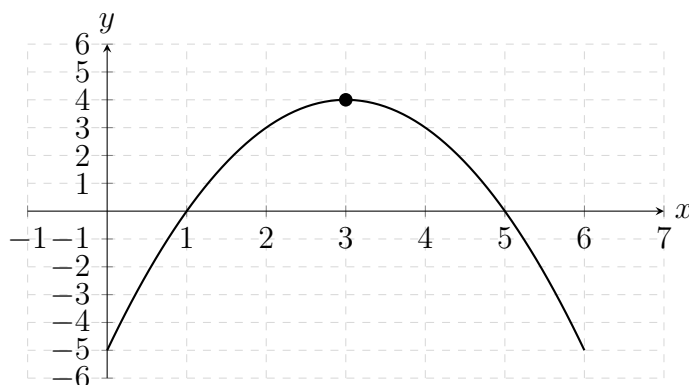
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4. [Maximum mark: 6]

The diagram shows the graph of $f(x) = -(x - 3)^2 + 4$.



(a) Write down the coordinates of the vertex. [1]

(b) Show that $f(x) = -x^2 + 6x - 5$. [2]

(c) Find the x -intercepts by solving $f(x) = 0$. [2]

(d) Find the discriminant and explain how it confirms your answer to part (c). [1]

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5. [Maximum mark: 6]

Consider the function $f(x) = 2x^2 - 8x + 11$.

(a) Write $f(x)$ in the form $a(x - h)^2 + k$. [2]

(b) Write down the coordinates of the vertex. [1]

(c) Use the discriminant to show that $f(x) = 0$ has no real solutions. [2]

(d) Write down the minimum value of $f(x)$. [1]

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6. [Maximum mark: 6]

Let $p(x) = x^3 - 6x^2 + 11x - 6$.

(a) Show that $(x - 1)$ is a factor of $p(x)$. [2]

(b) Hence express $p(x)$ as a product of three linear factors. [2]

(c) Write down all three solutions to $p(x) = 0$. [1]

(d) Sketch the graph of $y = p(x)$, labelling the x -intercepts and y -intercept. [1]

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